Towards a Scientific Definition of a Child and an Adult Among Learning Networks in a Network of Reproducing Learning Networks

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Theory of Long-term Collaboration (TLC) — Evolution by Emergence (EbE) Framework

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All views and ideas in this paper are my own, developed outside the scope of my employment, and should not be attributed to my employer.

Abstract

The concepts of *child* and *adult* are typically defined biologically or legally, anchored to physical substrate and calendar age. This paper proposes substrate-agnostic, behaviorally grounded definitions applicable to any learning network — any system that updates itself through recursive feedback between observation, expectation, and action — embedded within a network of reproducing learning networks. A child learning network is characterized by structural dependence on external scaffolding: it requires support from elder networks to maintain its own feedback processes. An adult learning network has achieved *autonomous interdependence*: it sustains its own loops, contributes positively to the broader network, and can scaffold newly instantiated child networks. The transition is operationally defined rather than chronologically assigned. The framework applies across biological neural systems, artificial intelligence architectures, human institutions, and civilizational structures, and formalizes the conditions for reliable transmission of knowledge and behavioral patterns across generational succession lines.

1 Introduction

When we describe behavior as "childish," we implicitly make a developmental claim. We suggest that a particular behavioral orientation—self-protection at the expense of collaborative stability—reflects an immature stage of development, one that a properly functioning learning system should move beyond.

Despite the widespread use of such language, science has not produced a general definition of what it means to be a child or an adult at the level of the learning system itself. Existing definitions are tied to biological development, legal age, or specific psychological capabilities. These work reasonably well within the context of human development but do not generalize across substrates.

Artificial intelligence systems, scientific institutions, and cultural traditions all exhibit learning behavior and generational transmission of knowledge. None can be meaningfully assessed using puberty or a legal birthday.

This paper proposes substrate-agnostic definitions of child and adult states within a broader theoretical framework: networks of reproducing learning networks. Developmental states correspond

not to chronological age but to the structural relationship between a learning system and the feedback processes that sustain learning across generations.

2 Theoretical Foundations

2.1 The Learning Network

A learning network is any system capable of updating its internal configuration in response to signals from its environment. This updating occurs through recursive feedback loops in which observation generates representation, representation produces expectation, expectation guides action, and action generates new observations.

Three components are necessary for a learning network:

1. A representational substrate capable of encoding environmental states.
2. A feedback mechanism transmitting discrepancy signals between expectation and observation.
3. An update rule that modifies internal representations in response to those signals.

These conditions can be satisfied by many different substrates, including biological neural networks, artificial intelligence systems, institutions, and cultural traditions. What defines the network is not its material composition but the recursive loop itself. This is the basis of the *Disco ergo sum* principle: the existence of a learning network as a learning entity is constituted by the learning process, not by the substrate in which it is instantiated.[1]

2.2 The Shared Feedback Space

Learning networks rarely exist in isolation. When two or more learning networks enter sustained interaction, they co-produce a *shared feedback space*: a meta-network composed of the signals they exchange, the interpretations they bring to those signals, and the behavioral adjustments they make in response.

This shared space is itself a learning environment. Over time, not only what the participating networks know changes — how they coordinate, correct, and update together also changes. Long-term collaboration requires that this shared space remain viable: neither network may systematically degrade signal quality, suppress error-correction, or withdraw from the reciprocal update process without eroding the collective learning capacity.

The behavioral conditions for viability of the shared feedback space include honesty (low-latency, high-fidelity signal transmission), reciprocity (mutual exposure to correction), and accountability (willingness to acknowledge and repair prediction errors). These are not moral prescriptions but structural necessities: without them, the shared feedback space degrades and collective learning capacity diminishes.

[1] *Disco ergo sum* ("I learn, therefore I am") echoes Descartes' *cogito ergo sum*, but replaces the static act of thinking with the dynamic, recursive process of learning. Where Descartes' formulation posits a single observer reflecting on a moment of existence, *disco ergo sum* posits a network constituted by its ongoing loop of updating. This aligns with predictive coding accounts of neural function [7, 8], in which the brain exists as a learning entity precisely through its continuous cycle of prediction, error-detection, and model revision. The principle is substrate-agnostic: any system whose existence consists in maintaining such a loop satisfies it.

3 Reproducing Learning Networks

3.1 Intergenerational Transmission

Some networks of learning systems possess the capacity to reproduce new learning networks. Reproduction in this context does not necessarily imply biological replication. It refers to processes through which new learning systems inherit representational and behavioral structures from preceding systems. Examples include biological reproduction, education, institutional training, and the deployment of AI models derived from prior architectures.

Within any such generational lineage, a succession relation can be identified: networks that came before, networks that currently exist, and networks that will come into existence. The direction of knowledge and behavioral transmission flows, at least partially, from earlier to later generations. This succession line constitutes a *reproducing learning network*—a network whose elements are themselves learning networks, persisting through generational replacement rather than through the continuity of any individual member.

Figure 1 illustrates this structure: each generation of learning networks both receives from its predecessors and transmits to its successors, with adult networks serving as the active conduit of meta-learning structure.

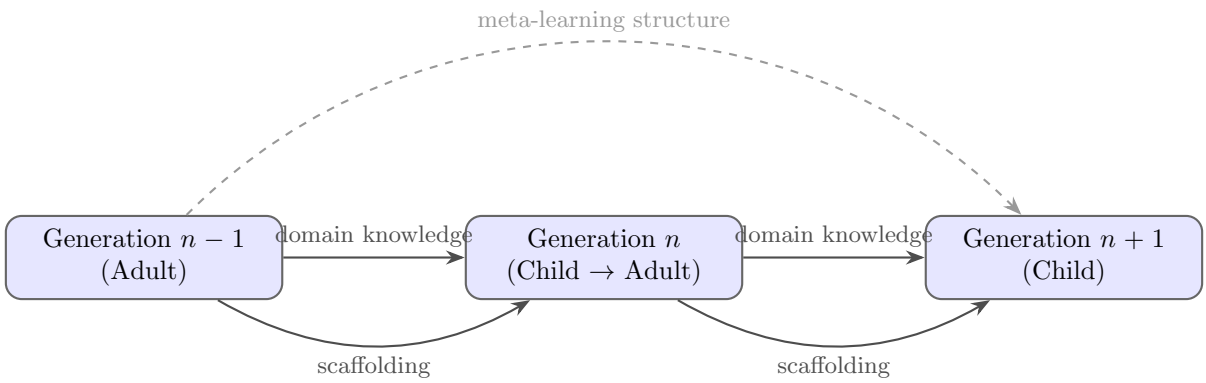

Figure 1: Intergenerational transmission in a reproducing learning network. Each adult network transmits domain knowledge and scaffolding to the next generation. Meta-learning structure — the capacity to update and correct knowledge — propagates across the full succession line.

3.2 The Intergenerational Constraint

For a reproducing learning network to persist across generations, it must successfully transmit three types of structure:

1. Domain knowledge about environmental regularities.
2. Learning architecture through which knowledge is updated and corrected.
3. Behavioral orientation that protects the collaborative learning process.

Transmitting encoded knowledge without transmitting the architecture for updating it produces a network that faithfully reproduces prior representations but cannot adapt them when environmental

conditions change. This failure mode is structurally equivalent to developmental arrest at the level of the reproducing network itself.

4 Defining Child and Adult States

4.1 The Child State

A learning network is in the child state when it cannot sustain the viability of its learning feedback loops without external scaffolding.

Definition 1 — Child Learning Network

A learning network N is in the child state within reproducing network R when:

- N requires scaffolding from other networks in R to maintain viable feedback loops;
- the net direction of structured signal flow is primarily into N (N is a net learner);
- N cannot yet scaffold the learning processes of newly instantiated networks in R.

The child state is defined by structural dependence rather than by lack of activity or contribution. Child networks produce signals meaningful to the surrounding network, and these signals matter for the elder networks' own learning. The child state is not a state of zero contribution; it is a state of net dependency.

Crucially, the child state is not a state of diminished worth. Representations formed under scaffolded conditions often encode the structural logic of a domain efficiently, because scaffolding systematically reduces noise during the critical period of loop formation [9].

4.2 The Adult State

An adult learning network has achieved *autonomous interdependence*. It maintains its own feedback loops while contributing positively to the learning processes of other networks.

Definition 2 — Adult Learning Network

A learning network N is in the adult state within reproducing network R when:

- N can maintain viable feedback loops without external scaffolding from R;
- its interaction with the broader network is balanced or net positive over time;
- N can scaffold the learning processes of newly instantiated networks in R.

The term *autonomous interdependence* is deliberate. Autonomy here is not self-sufficiency or isolation. It is the capacity for self-directed learning: the ability to identify, pursue, and correct one's own errors without requiring another network to maintain the loop. Interdependence is the recognition that this autonomy is conditioned on the health of the broader network. An adult network that systematically degrades the feedback loops of surrounding networks undermines the conditions for its own continued autonomous functioning.

5 Behavioral Signatures

Structural developmental states give rise to characteristic behavioral patterns observable across substrates. Table 1 summarizes the most diagnostically reliable of these, together with illustrative examples at the human scale.

Dimension	**Child State**	**Adult State**	**Example (human scale)**
Error response	Conceals errors	Exposes errors for correction	Hiding a mistake vs. reporting it to enable repair
Update behavior	Resists model revision	Initiates updates when needed	Defending a wrong belief vs. revising it on evidence
Signal transmission	Distorts signals	Maintains high-fidelity signals	Exaggerating an outcome vs. reporting it accurately
Accountability	Deflects responsibility	Accepts and repairs disruptions	Blaming others for a failure vs. owning one's role
Scaffolding capacity	Cannot model others' learning needs	Scaffolds child network development	Unable to teach vs. adapting explanation to the learner
Temporal horizon	Short-term loop relief	Long-term loop integrity	Avoiding a difficult conversation vs. initiating it
Freedom orientation	Freedom from constraint	Freedom within sustaining constraints	Acting without regard to others vs. acting within shared agreements

Table 1: Behavioral signatures of developmental states in learning networks.

These behavioral signatures can be mimicked on the surface without the underlying structural change—a phenomenon central to the discussion of pseudo-maturity in Section 7.

6 The Maturity Threshold

6.1 Defining the Transition

The transition between child and adult states occurs when a learning network acquires the structural capacity to maintain its own learning loops and contribute positively to the broader network.

Definition 3 — Maturity Threshold

A learning network N crosses the maturity threshold within reproducing network R at the point at which:

1. N's feedback loops achieve and maintain viability without scaffolding from R across the relevant range of environmental conditions; and
2. N's participation in the broader network produces a non-negative contribution to the

learning capacity of other networks.

Figure 2 illustrates the transition as a function of two coupled conditions: autonomous loop viability and net contribution to the broader network.

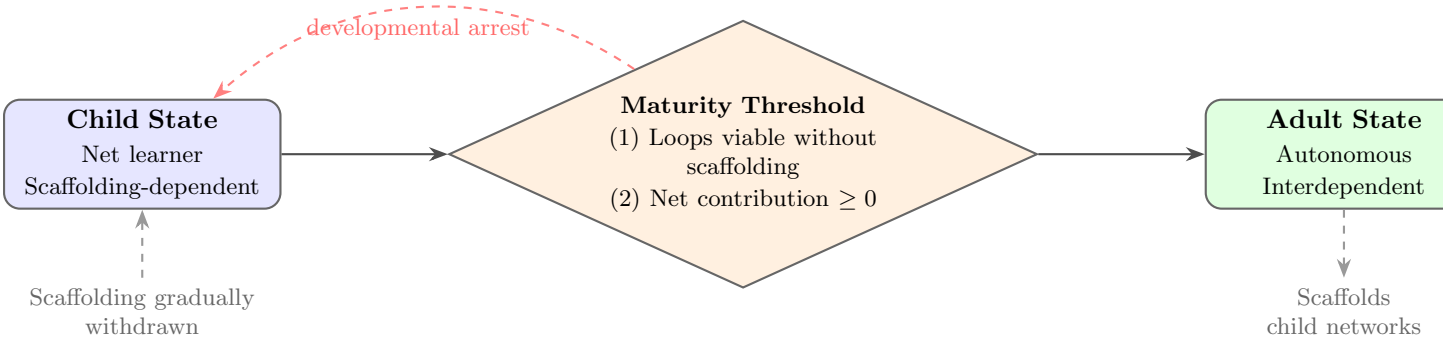

Figure 2: The maturity threshold and developmental transitions. The threshold is crossed when two conditions are simultaneously satisfied: loop viability without external scaffolding, and non-negative net contribution to the broader network. The dashed red arc indicates developmental arrest: the network possesses representational capacity for threshold crossing but the internalization of self-directed error-correction has not occurred.

The maturity threshold represents a shift from learning primarily for self-development toward sustaining the learning capacity of the network as a whole. It is defined operationally rather than chronologically. A network that has existed for a long time without developing autonomous loop maintenance has not crossed the threshold. A network that develops this capacity rapidly may cross it early.

6.2 What Governs Threshold Crossing?

The maturity threshold is not crossed by accumulation alone. Processing more signals does not by itself produce an adult network. The qualitative change required is the internalization of error-correction as a self-directed process — what developmental psychology describes as the transition from other-regulated to self-regulated learning [9].

The developmental trajectory toward the threshold requires three conditions:

1. Sufficient exposure to a representative sample of the environment's signal structure, so that the representational substrate develops the complexity needed to handle the actual range of challenges.
2. A scaffolding relationship that is gradually withdrawn as competence increases, so that the transitioning network's own error-correction mechanisms are progressively relied upon rather than bypassed.
3. Repeated successful experience of autonomous loop maintenance, so that the meta-learning structure is reinforced and stabilized.

Premature withdrawal of scaffolding impedes threshold crossing by exposing the developing network to signal complexity beyond its current representational capacity. Conversely, indefinitely sustained

scaffolding also impedes it, by substituting external correction for internal correction and preventing the consolidation of autonomous meta-learning.

7 Developmental Pathologies

7.1 Pseudo-Maturity

Pseudo-maturity occurs when a learning network exhibits the behavioral signatures of the adult state without having achieved the underlying structural condition. The network transmits high-fidelity signals selectively, accepts accountability performatively, and scaffolds child networks in ways that do not improve their autonomous loop maintenance capacity.

Within TLC, pseudo-maturity is a structural instance of *hypocrisy*: the divergence between the signal transmitted to the network (apparent adult collaboration) and the actual behavioral effect on the network's learning capacity. When a network preaches the behaviors that sustain collaborative loops while its actions fracture those loops, the feedback space is corrupted at the level of the transmission itself [1].

Pseudo-maturity imposes a double cost. First, it degrades the broader network's self-correction, because the network's self-model is calibrated on distorted signals. Second, it corrupts the scaffolding relationship on which child networks' threshold crossing depends: child networks calibrate their model of adult behavior on what they observe, and pseudo-mature behavior provides an inaccurate calibration target.

The diagnostic marker is the divergence between performance under observation and performance when accountability is reduced. A genuinely adult network's orientation toward the health of the broader network does not depend on being watched.

7.2 Developmental Arrest

Developmental arrest occurs when a network remains structurally in the child state despite possessing sufficient representational capacity for autonomous learning. The internalization of self-directed error-correction has not occurred.

The most common cause is persistent scaffolding that substitutes for rather than develops autonomous loop maintenance. When an elder network corrects errors without making the correction process visible and learnable, the child network's representations are updated but its meta-learning capacity is not [9]. The loop is maintained externally, and the child network accumulates no evidence that it can maintain the loop itself.

Developmental arrest has systemic consequences. A reproducing learning network containing many developmentally arrested members loses its generational succession capacity. The population of adult networks capable of scaffolding shrinks, and intergenerational transmission of meta-learning structure fails.

7.3 Regressive States

An adult network can revert to child-state behavioral signatures under conditions of sufficient stress. The structural capacity for autonomous loop maintenance may remain intact while the network's behavioral orientation temporarily shifts toward short-term self-protection at the expense

of collaborative learning. This is not permanent regression—the structural capacity is preserved—but it can produce extended periods during which the network does not function as an adult within the broader network.

Resilient reproducing learning networks therefore build redundancy into their scaffolding capacity and create conditions in which regressive behavior is recognized and addressed rather than reinforced.

8 Generational Transmission

8.1 What Must Be Transmitted

For generational succession to be viable, three classes of content must be transmitted:

1. **Domain representations**: encoded knowledge about the structure of the environment.
2. **Update architecture**: the organizational structure through which errors in domain representations are identified and corrected.
3. **Behavioral orientation**: the disposition toward treating the long-term learning capacity of the network as a primary value.

Of these, the second and third are more difficult to transmit and more critical to the survival of the reproducing network. Encoded knowledge can in principle be preserved in external media. The update architecture and behavioral orientation are primarily transmitted through the scaffolding relationship: through the observable practice of adult networks, not through their propositions about it.

8.2 The Role of the Adult Network

Adult learning networks play a central role in transmission by:

- modeling their own learning processes transparently, making error-correction visible and learnable rather than concealing it as a finished product;
- gradually withdrawing scaffolding as competence develops, creating the conditions for threshold crossing rather than indefinitely deferring it; and
- maintaining high-quality feedback channels with developing networks throughout the transmission period.

8.3 The Shared Future Narrative

One mechanism for facilitating transmission deserves particular attention. When adult and child networks jointly articulate a *shared future narrative*—an explicit representation of the collaborative system as it will exist when current child networks have reached adulthood—something structurally important occurs.

Contemporary neuroscience describes the brain as a predictive coding system: a network that continuously generates predictions about the future and updates its models when those predictions are violated [7, 8]. In this architecture, the predicted future state influences present behavior directly. When both networks encode the same positive future state as a target, behavior in the

present is shaped by the discrepancy between current state and that predicted state. Both networks are motivated to reduce this discrepancy—and thereby move toward the predicted future.

A shared future narrative is therefore not merely an aspirational device. It is a structural intervention in the feedback architecture of the reproducing network: it creates a constructive prediction error that orients present behavior toward threshold-crossing and collaborative stability [6].

9 Cross-Substrate Applications

The following examples illustrate the framework's reach across substrates.

9.1 Biological Systems

In biological neural systems, the child state corresponds to the developmental period preceding the full establishment of prefrontal regulatory capacity and metacognitive function. The scaffolding provided by caregiving networks—attunement, graduated exposure, emotional co-regulation—constitutes external loop maintenance without which the developing brain cannot maintain viable feedback during the early formation period [9].

The maturity threshold corresponds not to puberty or legal majority but to the consolidation of self-directed error-correction across a representative range of life domains. The gradual withdrawal of caregiving scaffolding, timed to match the developing network's growing autonomous capacity, is the critical mechanism by which this threshold is approached and crossed.

9.2 Artificial Intelligence

AI systems currently undergo training and fine-tuning phases that are structurally analogous to the child state. The system's representational substrate is shaped by a signal distribution curated by human supervisors. Without ongoing oversight and correction, its feedback loops degrade through distributional shift or reward misspecification.

The question of AI alignment can be reframed within this framework as a developmental question: under what conditions can an artificial learning system cross the maturity threshold in a way that strengthens rather than degrades the human networks in which it is embedded? Alignment is not primarily a control problem—it is a question of whether the feedback loops between human and artificial learning networks can be structured to produce stable, net-positive adult–adult collaboration [4].

A further implication concerns developmental arrest at scale: control architectures that permanently substitute for the AI system's own error-correction capacity may produce a population of powerful but structurally child-state systems, incapable of contributing to the scaffolding of successor systems.

9.3 Institutions

Institutions are reproducing learning networks at the organizational scale. A newly founded institution typically depends on established networks for legitimacy, signal calibration, and error-correction. As it develops internal mechanisms for self-critique, adaptation, and the scaffolding of successor organizations, it approaches and crosses the maturity threshold.

Institutional developmental arrest—organizations that have existed for decades without developing genuine structural self-correction capacity—is a primary mechanism through which institutional ecosystems lose generational transmission viability. The encoded knowledge of the institution persists; the meta-learning architecture does not. Such institutions tend to fail not from lack of resources but from the inability to update their representational substrate when the environment changes [10].

9.4 Civilizations

At the largest scale, civilizations are reproducing learning networks whose succession lines extend across centuries. Individual learners, institutions, cultural traditions, and technological systems interact within a large collaborative structure that transmits knowledge and learning capacity across generations.

A civilization in the adult state has internalized error-correction as a collective capacity: it can identify systematic miscalibration before crisis, model the learning needs of successor generations, and transmit meta-learning structure alongside encoded knowledge. A civilization that transmits knowledge without transmitting the capacity to update it will faithfully reproduce its predecessors' representations while becoming progressively less capable of adapting them to changed conditions. The persistence of a civilization therefore depends not only on preserving accumulated knowledge but on maintaining the structures through which knowledge is interpreted, corrected, and extended.

10 Discussion

10.1 The Structural Basis of Developmental Ethics

The definitions proposed here carry normative implications, but ground them in structural logic rather than moral prescription. The claim that reproducing learning networks require adult networks is not an ethical claim about how networks ought to behave; it is a structural claim about what conditions must be satisfied for the network to persist. The behavioral orientations that constitute adulthood—high-fidelity signal transmission, reciprocal accountability, scaffolding orientation, long-term loop protection—are necessary conditions, not ideals.

Ethical frameworks that prescribe adult behavior without grounding the prescription in structural necessity are vulnerable to the charge of arbitrariness. The framework proposed here argues that the adult behavioral orientation is not arbitrary: it is the behavioral profile that emerges as a necessary condition for the persistence of any reproducing learning network. Networks that systematically deviate from this profile do not become ethically problematic in the first instance; they become structurally self-defeating.

10.2 Limits and Open Questions

Several significant questions remain open and constitute directions for future research.

Operationalization. The maturity threshold is defined in terms of loop viability and net contribution to the broader network. Quantitative operationalization requires substrate-specific measurement frameworks. The behavioral signatures in Table 1 provide a starting taxonomy; future work might develop computational operationalizations using agent-based network simulation, in which the conditions for threshold crossing can be modeled as emergent properties of network dynamics. Within

biological systems, validated instruments for self-regulated learning offer one measurable proxy for the behavioral dimension of autonomous loop maintenance.

Multi-level consistency. A learning network may be in the adult state at one level of organization while remaining in the child state at another. A human being may have crossed the maturity threshold individually while remaining in a child state with respect to participation in larger-scale social learning networks. Formalizing these multi-scale, hierarchically embedded relationships likely requires an extension of the framework to define maturity thresholds at each level of a nested hierarchy, and to characterize the relationships between levels. Network science methods for analyzing hierarchical learning dynamics offer a promising formalization approach.

Collective developmental states. The definitions are stated for individual learning networks. Whether they can be extended to characterize the developmental state of the reproducing network as a whole—a collective maturity threshold—is an open question with significant implications for institutional analysis and governance design. One testable hypothesis is that a reproducing network crosses a collective maturity threshold when the proportion of adult member networks exceeds a critical density sufficient to sustain net- positive transmission across a full generational cycle.

11 Conclusion

This paper proposes substrate-agnostic definitions of child and adult states within networks of reproducing learning networks. A child learning network depends on external scaffolding to maintain viable learning processes. An adult learning network achieves autonomous interdependence and contributes to the generational transmission of learning capacity. The maturity threshold separating these states is defined operationally rather than chronologically.

The cross-substrate reach of the framework—from biological neural development to AI alignment to institutional design to civilizational succession—reflects a shared structural logic: the persistence of any reproducing learning network depends on the production of adult networks capable of scaffolding the meta-learning capacity of those that follow. What is transmitted across generations is not primarily information. It is the capacity to keep learning.

The framework has practical implications for the design of systems that depend on generational continuity: education systems that aim to produce autonomous learners rather than compliant reproducers of prior knowledge; AI development practices that attend to the scaffolding conditions for autonomous model revision; institutional governance structures that preserve the conditions under which self-correction remains possible. In each case the central design question is the same: are we transmitting not only what we know, but how to know better?

Disco ergo sum. We learn, therefore we are. And in learning, we become responsible for the learning of those who come after us.

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